FIT5230 Malicious AI

Deepfakes II

Deepfakes vs Anti-Deepfakes

- Deepfakes
  - AI attacks Security property of INT

- Anti-Deepfakes
  - prevent deepfakes: can't, just like crypto also can't
  - detect deepfakes:
    - similar to
      - crypto:
        - checkcode/signature valid for the media?
      - watermarking:
        - valid embedded watermark can still be extracted?

How digital signature works in cybersecurity:

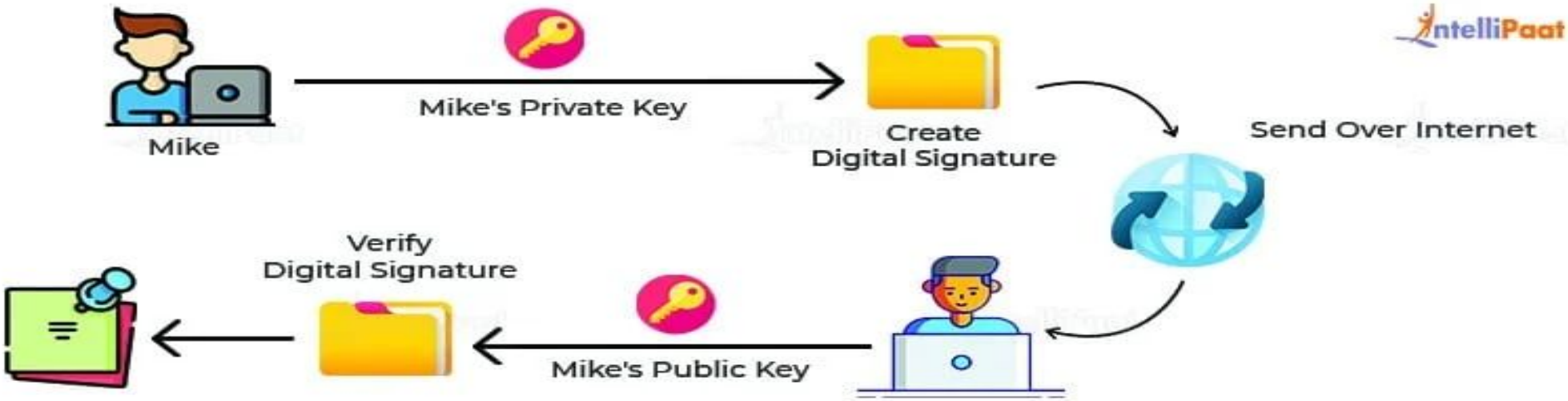
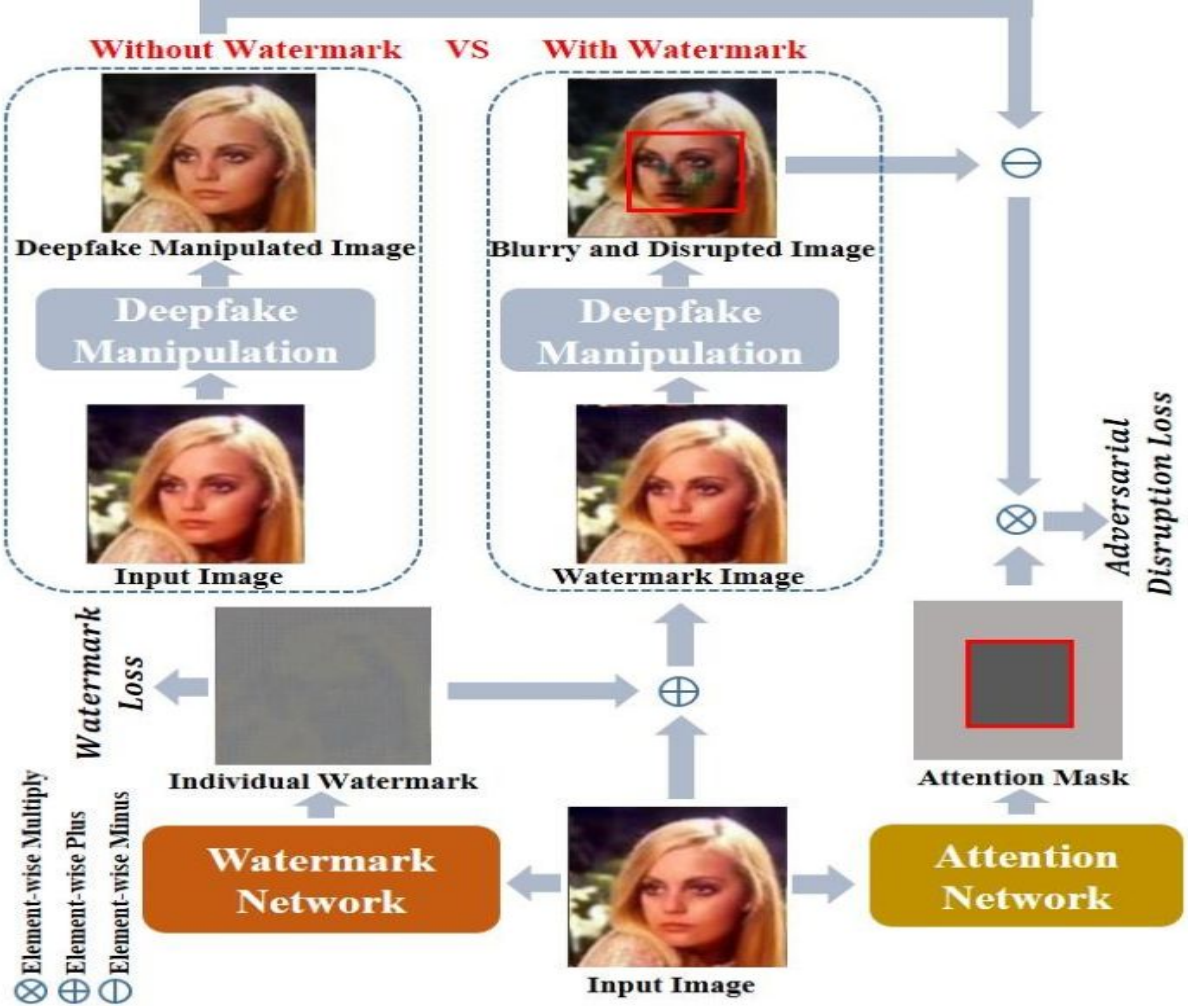


Image source

How anti-deepfakes works:



Lv, L. (2021). Smart Watermark to Defend against Deepfake Image Manipulation. *2021 IEEE 6th International Conference on Computer and Communication Systems (ICCCS)*, 380-384.

How to distinguish between Real & Fake?

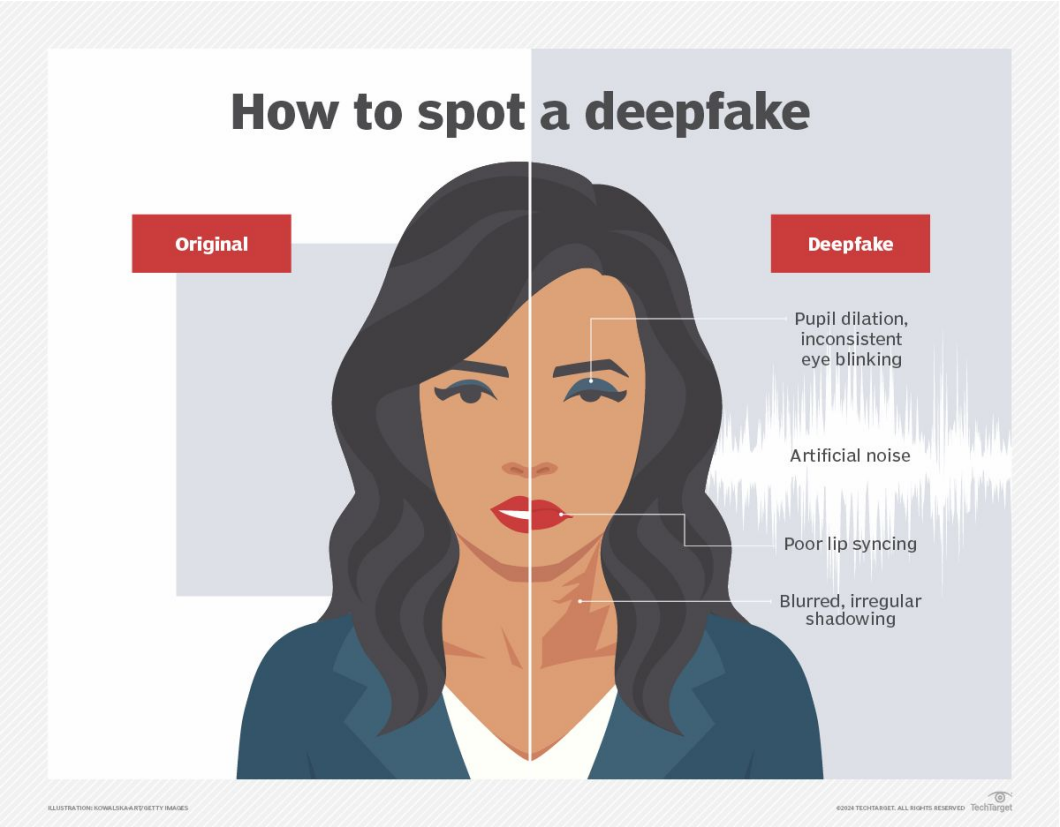
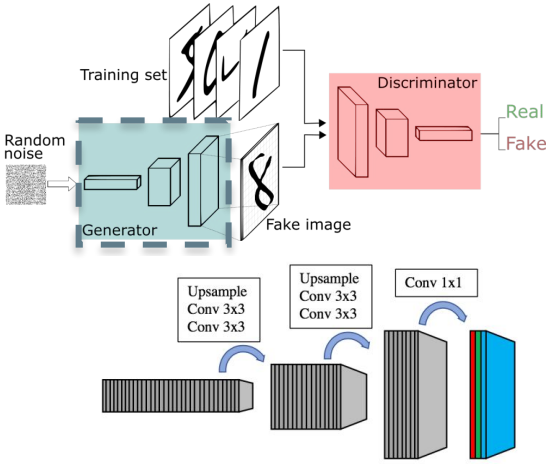


Image source

Artefacts of Deepfake Generators

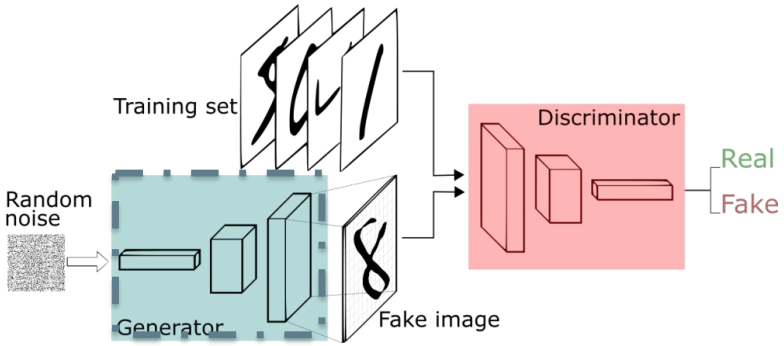


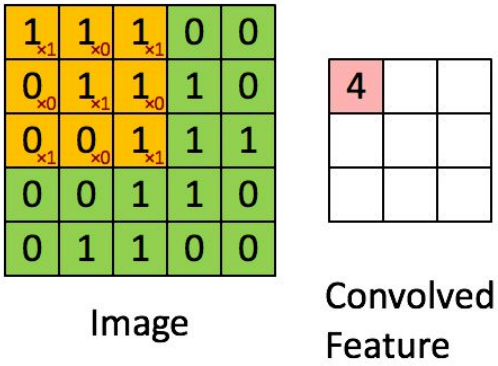


- Real images: Formed by actual cameras and sensors, based on how light is captured.
- Deepfakes: Generated by AI through upsampling & convolution → can leave blocky artefacts that look unnatural.

https://arxiv.org/pdf/1812.08247.pdf

Convolution (Downsampling)

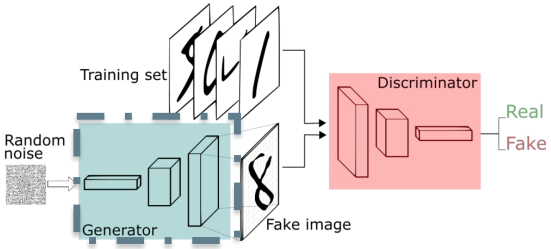


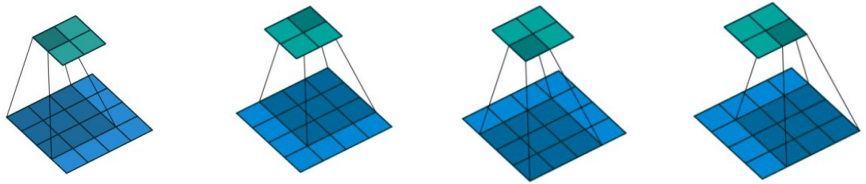


- focus on spatial patterns/features in input data
- kernel, filter: 3×3, each element is multiplier
- imagine sliding the kernel over each pixel of the 2D image

https://github.com/vdumoulin/conv_arithmetic

Convolution (Downsampling)



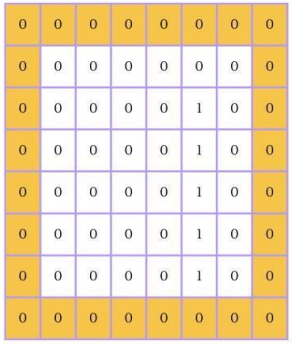


- convolution
  - blue: input
  - cyan: output after convolution
  - dark blue: sliding the kernel across the input

  - example: no padding, no strides (skips)

https://github.com/vdumoulin/conv_arithmetic

Convolution: Padding



Zero Padding

Image source

Replication Padding

5	5	0	8	7	8	1	1
5	**5**	**0**	**8**	**7**	**8**	**1**	1
1	**1**	**9**	**5**	**0**	**7**	**7**	7
6	**6**	**0**	**2**	**4**	**6**	**6**	6
9	**9**	**7**	**6**	**6**	**8**	**4**	4
8	**8**	**3**	**8**	**5**	**1**	**3**	3
7	**7**	**2**	**7**	**0**	**1**	**0**	0
7	7	2	7	0	1	0	0

Reflection Padding

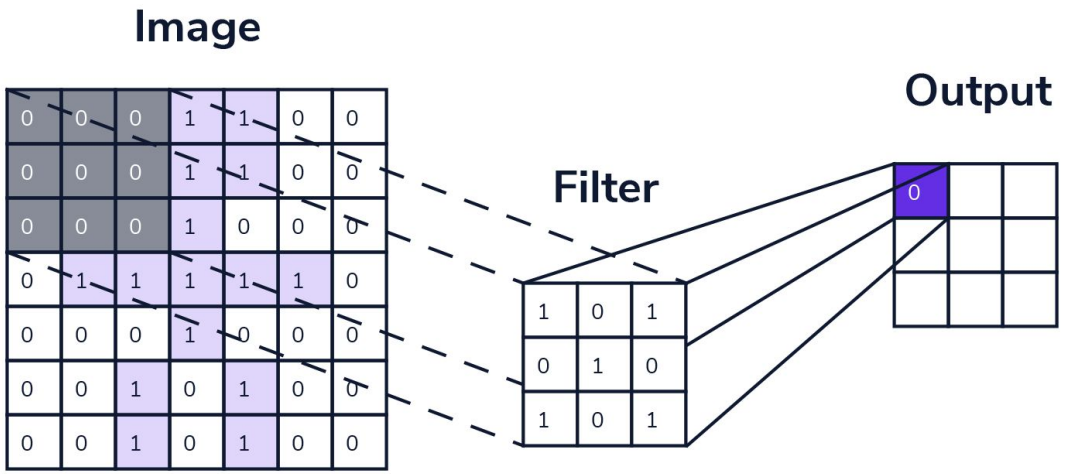
9	1	9	5	0	7	7	7
0	**5**	**0**	**8**	**7**	**8**	**1**	8
9	**1**	**9**	**5**	**0**	**7**	**7**	7
0	**6**	**0**	**2**	**4**	**6**	**6**	6
7	**9**	**7**	**6**	**6**	**8**	**4**	8
3	**8**	**3**	**8**	**5**	**1**	**3**	1
2	**7**	**2**	**7**	**0**	**1**	**0**	1
3	8	3	8	5	1	3	1

Circular Padding

0	7	2	7	0	1	0	7
1	**5**	**0**	**8**	**7**	**8**	**1**	5
7	**1**	**9**	**5**	**0**	**7**	**7**	1
6	**6**	**0**	**2**	**4**	**6**	**6**	6
4	**9**	**7**	**6**	**6**	**8**	**4**	9
3	**8**	**3**	**8**	**5**	**1**	**3**	8
0	**7**	**2**	**7**	**0**	**1**	**0**	7
1	5	0	8	7	8	1	5

Image source

Convolution - Stride



Stride = 2

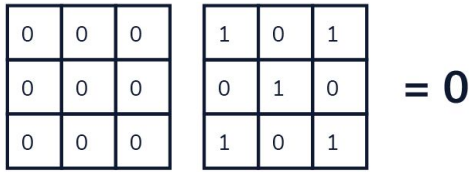


Image source

Convolution: Pooling

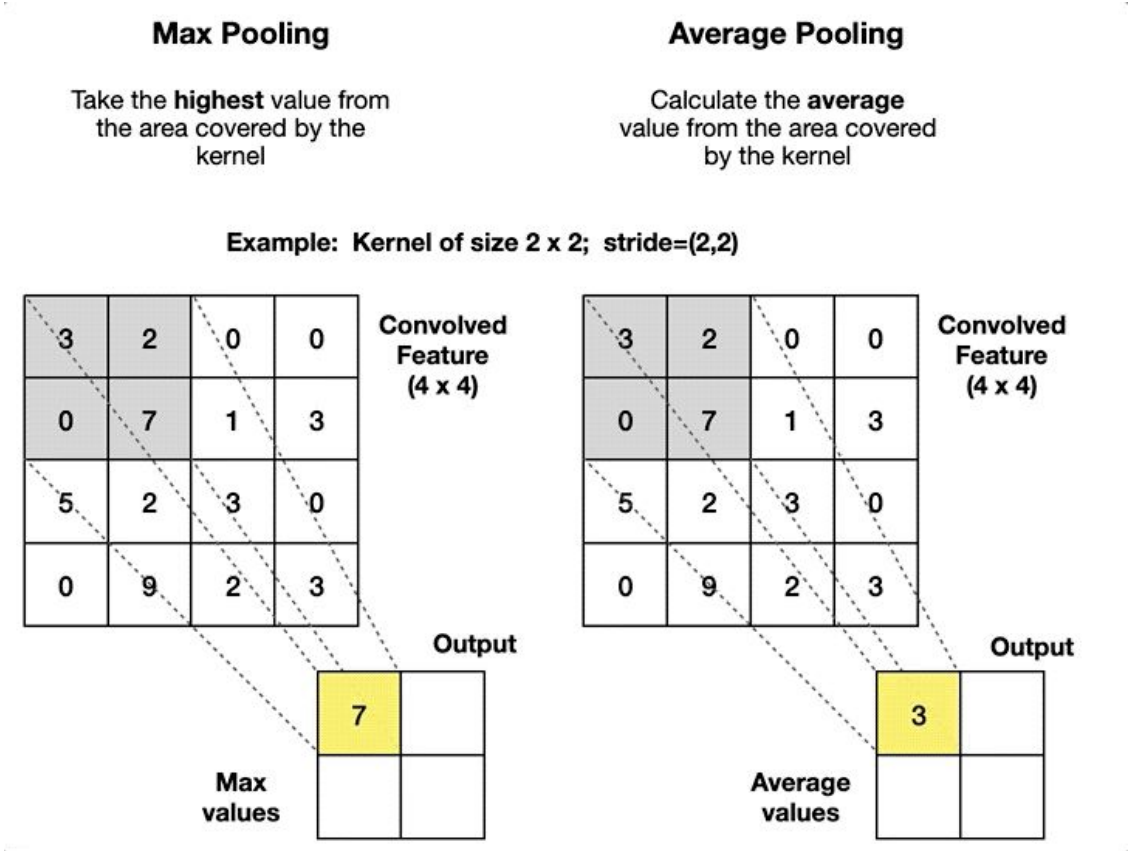
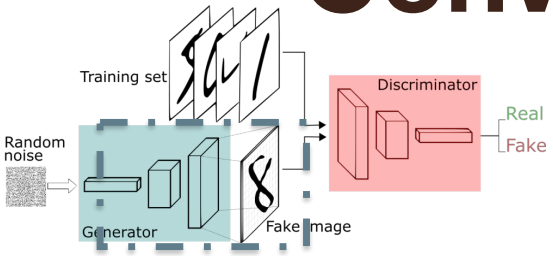
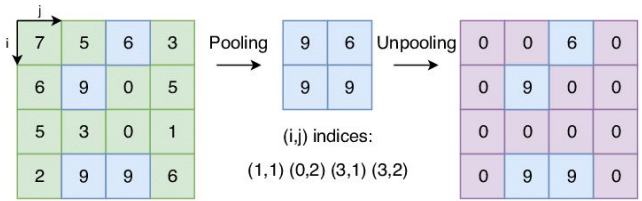


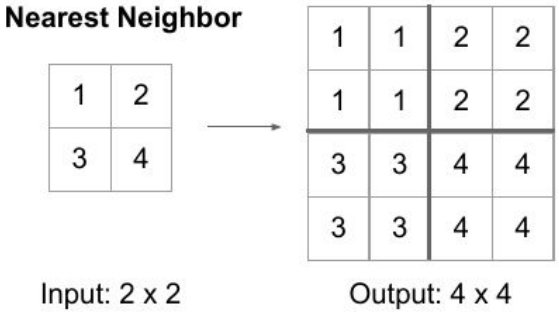
Image source

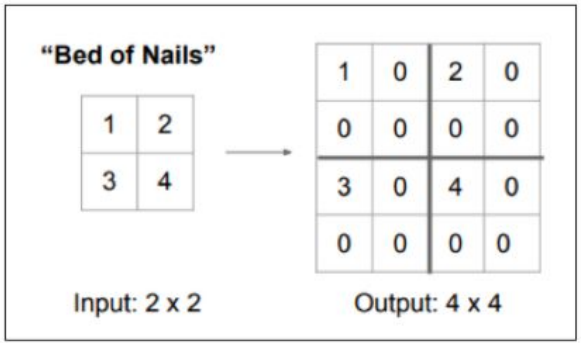
Convolution: Upsampling



- basically the reverse of downsampling, increase size
- e.g. unpooling







https://kharshit.github.io/blog/2019/02/15/autoencoder-downsampling-and-upsampling
https://www.researchgate.net/figure/Example-of-an-unpooling-process-Indices-of-max-pooling-are-kept-up-and-reused-to_fig4_335754645

MesoNet

MesoNet: a Compact Facial Video Forgery Detection Network

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Abstract

This paper presents a method to automatically and efficiently detect face tampering in videos, and particularly focuses on two recent techniques used to generate hyper-realistic forged videos: Deepfake and Face2Face. Traditional image forensics techniques are usually not well suited to videos due to the compression that strongly degrades the data. Thus, this paper follows a deep learning approach and presents two networks, both with a low number of layers to focus on the mesoscopic properties of images. We evaluate those fast networks on both an existing dataset and

falsified video raises more and more concerns. While significant improvements have been made for image forgery detection, digital video falsification detection still remains a difficult task. Indeed, most methods used with images can not be directly extended to videos, which is mainly due to the strong degradation of the frames after video compression. Current video forensic studies [16] mainly focus on the video re-encoding [28] and video recapture [29, 15], however video edition is still challenging to detect.

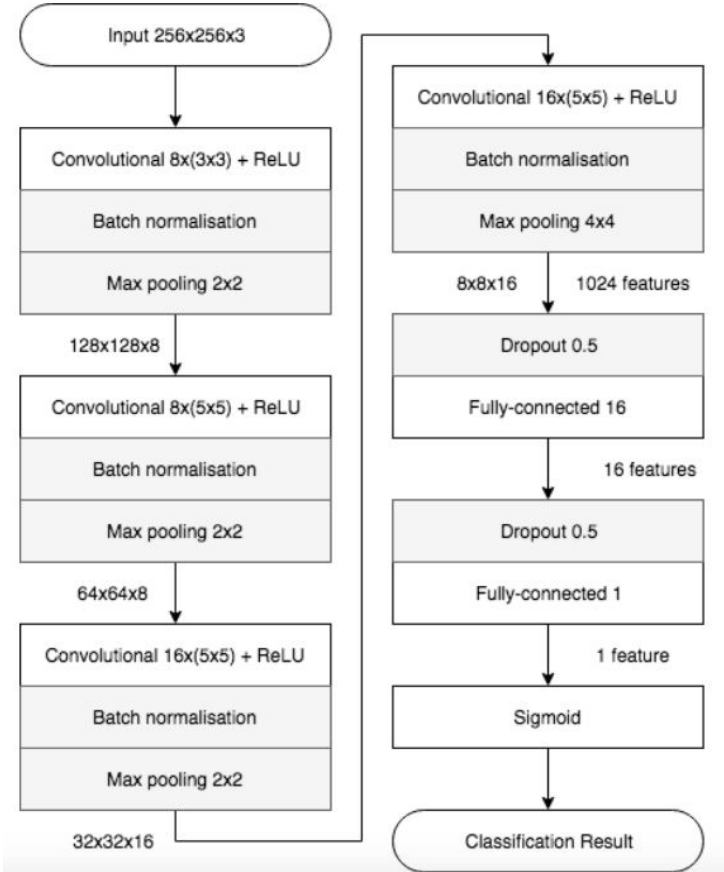
For the last years, deep learning methods has been successfully employed for digital image forensics. Amongst others, Barni et al. [2] use deep learning to locally detect

@ WIFS 2018

MesoNet

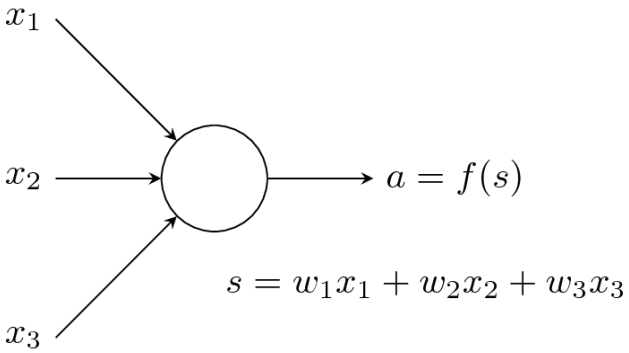
- Microscopic: based on noise
- Macroscopic: higher semantics
  - e.g. image-wise
- Mesoscopic: in between

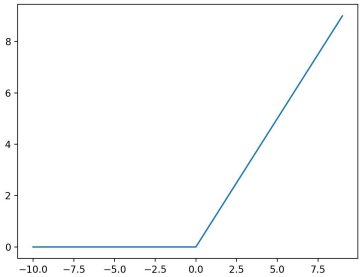
- Meso-4 (CNN)
  - convolution
  - rectified linear unit (ReLU)
  - batch normalisation
  - max pooling
  - dropout
  - fully connected
  - sigmoid



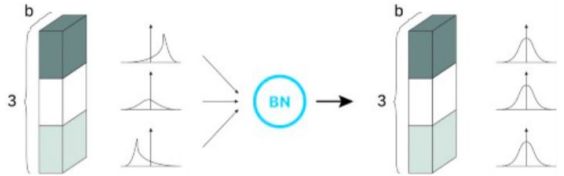
MesoNet

- @ neural network node:





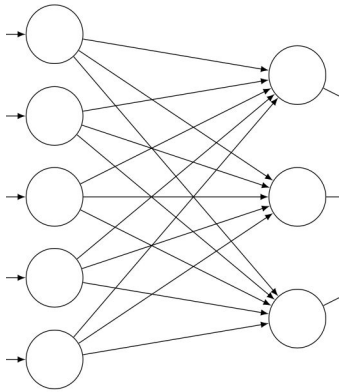
- Rectified linear unit (ReLU)
  - a type of activation function f( )
  - pass through only if +ve



- batch normalisation
  - subtract batch mean μ, divide by batch standard deviation σ

MesoNet

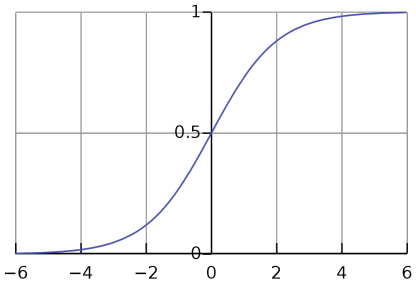
- dropout
  - strategy for fully connected layer
  - to regularize (slight modifications to learning for better generalization, prevent overfitting) the output
  - to simulate effect of simultaneously training many neural networks with different architectures

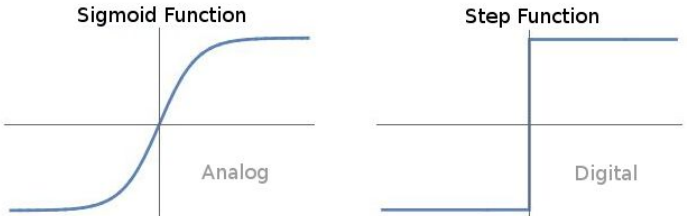


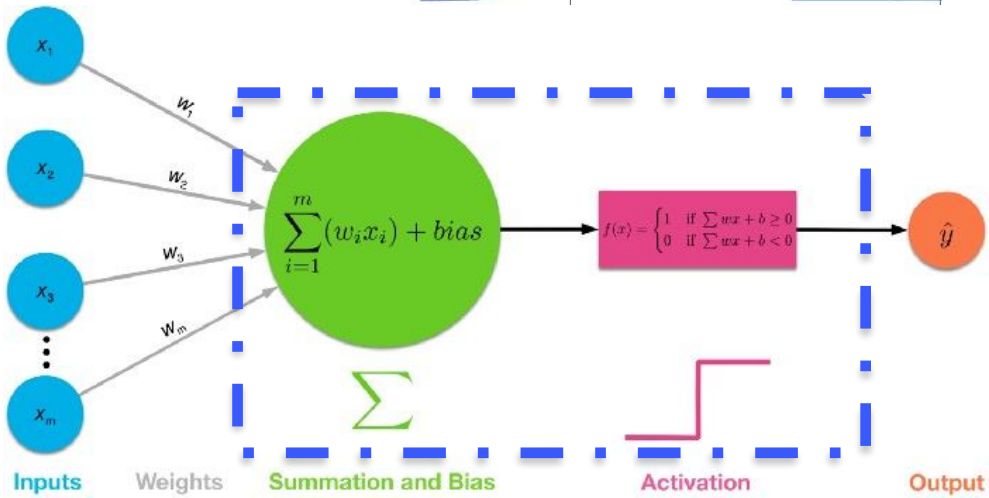
- fully connected

MesoNet

- sigmoid: non-linear activation function







EnsembleNet

Video Face Manipulation Detection Through Ensemble of CNNs

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***Abstract*—In the last few years, several techniques for facial manipulation in videos have been successfully developed and made available to the masses (i.e., FaceSwap, deepfake, etc.). These methods enable anyone to easily edit faces in video sequences with incredibly realistic results and a very little effort. Despite the usefulness of these tools in many fields, if used maliciously, they can have a significantly bad impact on society (e.g., fake news spreading, cyber bullying through fake revenge porn). The ability of objectively detecting whether a face has been manipulated in a video sequence is then a task of utmost importance.**

In this paper, we tackle the problem of face manipulation de-

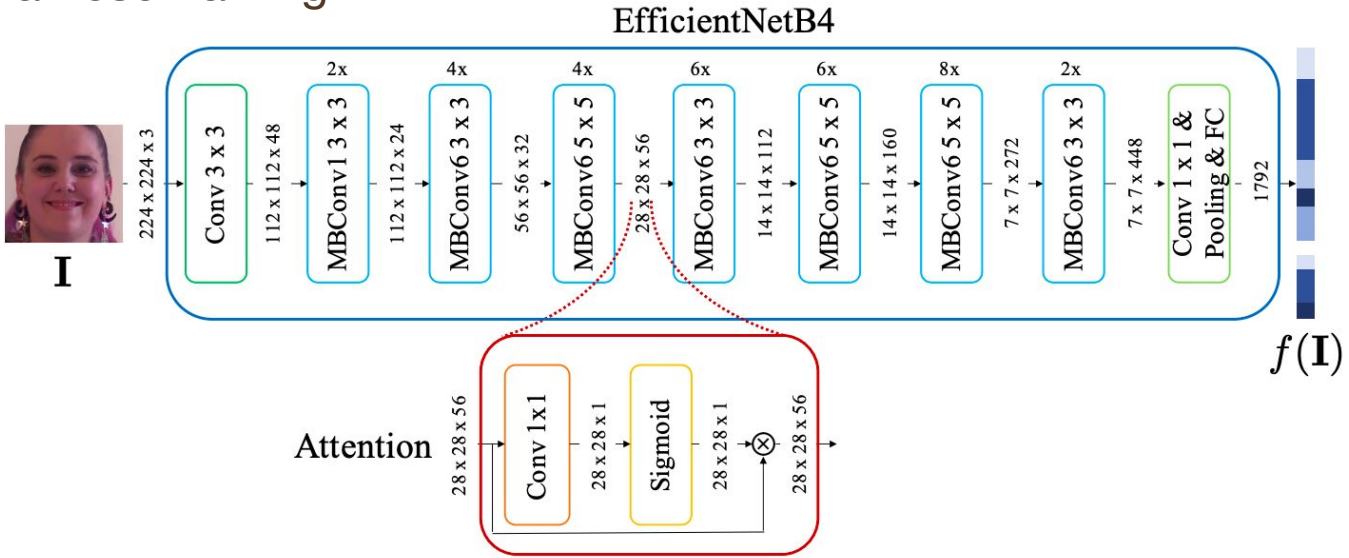
to new artistic possibilities (e.g., movie making, visual effect, visual arts, etc.). On the other hand, unfortunately, it also eases the generation of video forgeries by malicious users.

Fake news spreading and revenge porn are just a few of the possible malicious applications of advanced facial manipulation technology in the wrong hands. As the distribution of these kinds of manipulated videos indubitably leads to serious and dangerous consequences (e.g., diminished trust in media, targeted opinion formation, cyber bullying, etc.), the ability of detecting whether a face has been manipulated in a video

@ ICPR 2020

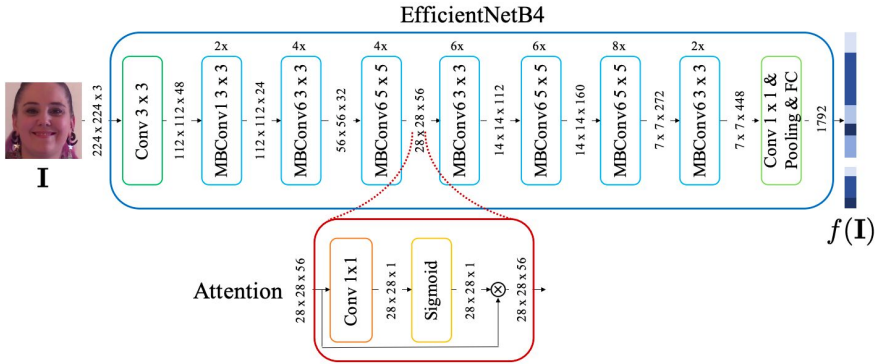
EnsembleNet

- ensemble of CNNs: +
  - attention layer
  - siamese training



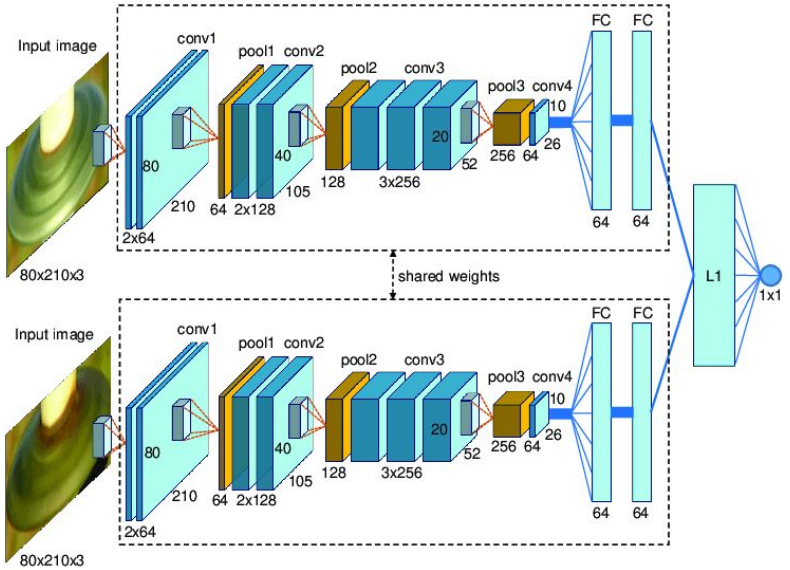
EnsembleNet

- ensemble of CNNs: +
  - attention layer
    - based on convolution & sigmoid activation, get attention map, multiplied ⊗ back into the feature map
    - learn long-range dependencies/relationships btw different points in time
    - infer which portion of video important for classification



EnsembleNet

- ensemble of CNNs: +
  - attention layer
  - siamese training
    - same network & weights simultaneously on two inputs



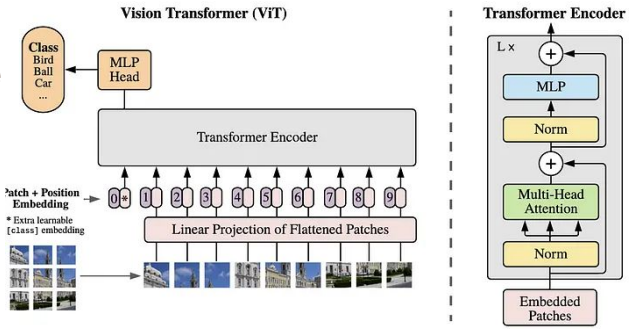
Vision Transformer (ViT)

- Deepfake artefacts are often subtle and can span non-contiguous regions (e.g., inconsistent lighting across the entire face, unnatural reflections in eyes, irregularities in hair boundaries).

What's wrong with CNN?

  - CNNs struggled to model global relationships between different parts of an image.
  - The self-attention mechanism of ViT inherently captures global dependencies across all image patches in a single layer.

- This allows ViT to detect inconsistencies in the global coherence of an image that might be missed by models focusing on local features.



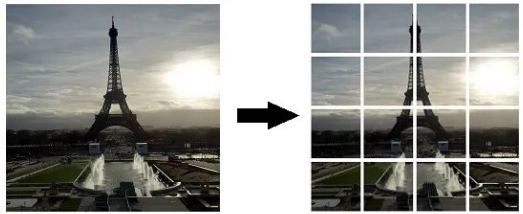
2010.11929v2.pdf (arxiv.org)

Vision Transformer (ViT)

How ViT works?

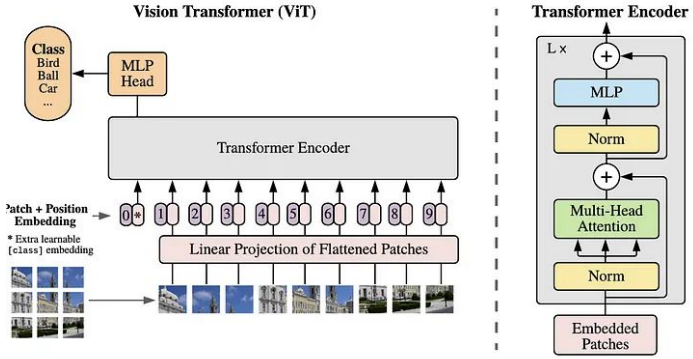
- Patchify the Image:
  - Split the input image into smaller, fixed-size non-overlapping patches.
  - Imagine an image of size 128×128 pixels.
  - If ViT splits the image into 16×16 pixels, this would create 64 patches
    - 128÷16=8 patches along width, and 8 patches along height, making 8×8=64 patches)

https://mrinath.medium.com/vit-part-1-patchify-images-using-pytorch-unfold-716cd4fd4ef6



Vision Transformer (ViT)

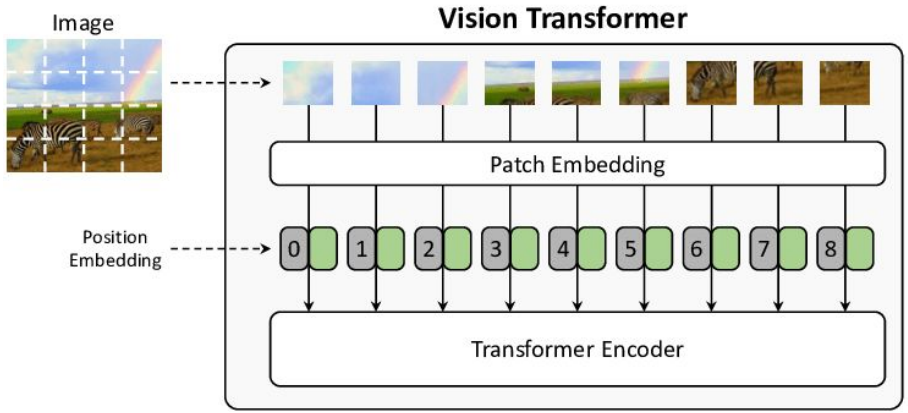
- Linear Projection:
  - Flatten each patch into a 1D vector (token).
    - 16×16×C patches are flattened into a single vector
    - conceptually like turning a mini image into a word in a sentence.
    - These flattened vectors are then projected into a consistent embedding dimension.



2010.11929v2.pdf (arxiv.org)

Vision Transformer (ViT)

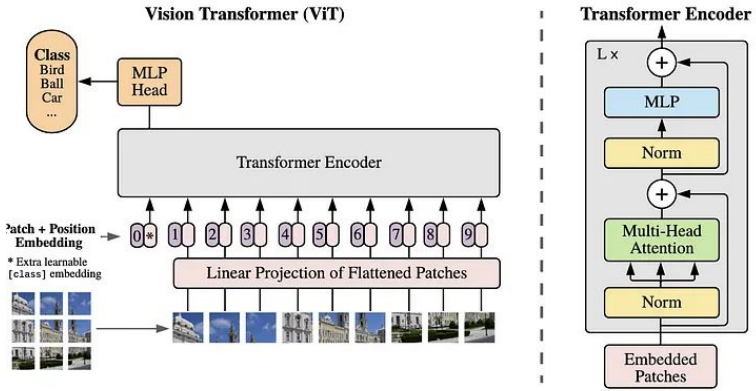
- Positional Encoding:
  - Learnable 1D positional embeddings are added to each patch token to explicitly inform the Transformer of **where** each patch is **located** in the original image, thereby preserving **crucial spatial information**.



https://www.researchgate.net/figure/sion-Transformer-encoding-The-image-is-split-into-fixed-size-patches-linearly-embedded_fig1_353284955

Vision Transformer (ViT)

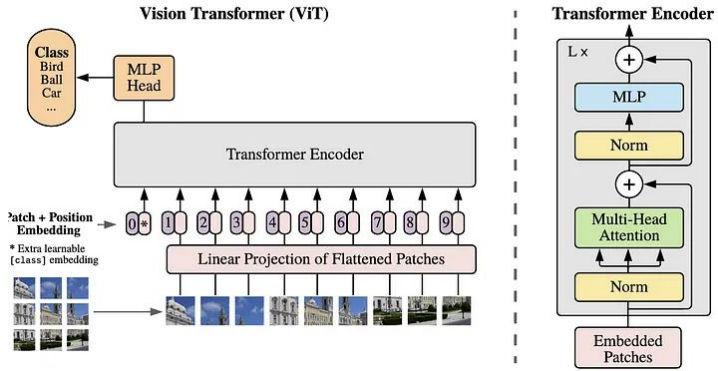
- Transformer Encoder:
  - Process the sequence of image tokens through Transformer layers.
  - Each patch is now represented as a **token**.
  - The token passes through the Transformer encoder.
  - The self-attention mechanism helps each patch learn about other patches in the image.



2010.11929v2.pdf (arxiv.org)

Vision Transformer (ViT)

- Classification Token:
  - After processing the image tokens, ViT uses a **special classification token**.
  - This token aggregates the **global information** from all patches and is passed through a simple neural network to make the **final classification** (e.g., identifying the object in the image).

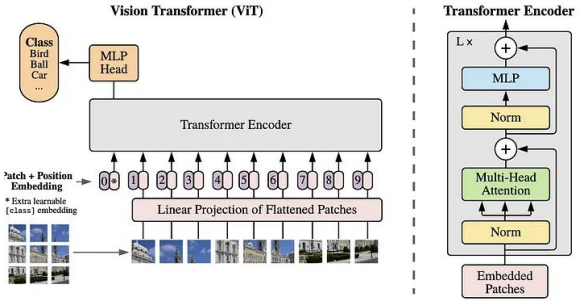


2010.11929v2.pdf (arxiv.org)

Vision Transformer (ViT)

Why ViT?

- Modern deepfakes aim to minimize localized "sharp" artefacts, spreading imperfections thinly across the image.
- By considering relationships between all patches, ViT can pick up on subtle, distributed statistical anomalies or unnatural correlations between seemingly disparate image regions that collectively signal manipulation.
- ViT has a weaker inductive bias, allowing it to learn more flexible and complex patterns directly from data. This can be beneficial for detecting unforeseen types of synthetic artefacts



2010.11929v2.pdf (arxiv.org)

Vision Transformer (ViT)

- **Temporal Consistency in Video Deepfakes:**
  - While single deepfake frames can be convincing, maintaining perfect temporal coherence across a video sequence is extremely difficult for generative models.
    - Natural blinking, consistent head movements, realistic facial expressions over time
  - ViT can be extended to video by creating tokens from image patches and treating frames as elements in a sequence.
    - Model learns spatiotemporal attention, capturing how features evolve not just across space (within a frame) but also across time (between frames).
    - Enables the detection of "temporal glitches" or unnatural motion patterns unique to deepfakes.

Anti-Deepfake Techniques

- MesoNet (CNN-based)
  - Lightweight model that looks at mesoscopic features
    - Fast, efficient and works well on images
  - BUT…
    - Limited ability to catch global inconsistencies

- EnsembleNet
  - Combines multiple CNNs
  - Adds attention layer to focus on important parts of face
  - Included siamese training to compare pairs of inputs
  - More robust than single CNN
  - Good at learning local and temporal features in video
  - BUT…
    - Computationally heavier than MesoNet

Anti-Deepfake Techniques

- Recurrent Neural Network
  - Designed for sequential data
    - time series, video frames, speech
  - Learns temporal dependencies like blinking, lip sync and head movement
    - Good at detecting temporal glitches across video frames
  - BUT…
    - Struggles with spatial details in single frames

Anti-Deepfake Techniques

- Vision Transformer (ViT)
  - Splits images into patches, processes with self-attention to capture global relationships.
  - Extended to video with spatiotemporal attention.
    - Excellent at spotting subtle, distributed anomalies across image/video
  - BUT…
    - Needs large datasets and heavy computation
    - Slower than CNNs.

Activity 1

What you need to do:

1. **Choose** one of the detection method:
   - MesoNet
   - EnsembleNet
   - Vision Transformer
   - Watermarking/cryptographic signatures

1. Imagine you are **designing** an anti-deepfake defense system for one of these organizations:
   - A news agency
   - A hospital
   - A social media platform

3. **Discuss** in your group:
   - What types of deepfakes are your organization most vulnerable to?
   - How does your assigned method help defend against them?
   - What are the weaknesses or limitations of your method?
   - Would you combine it with other strategies to make it stronger?

Provide your answer here: https://shorturl.at/zUzzJ

Ideal Anti-Deepfake

- Preventive Layer → Watermarking & Cryptographic Signatures
  - Encourage content authentication at the source: news agencies, hospitals, and official institutions
    - embed invisible digital watermarks or cryptographic signatures in all original media
  - Platforms prioritize or “verify” signed content.



Image source

Ideal Anti-Deepfake

- Human-in-the-Loop Verification
  - For cases where AI gives low-confidence or conflicting results, human experts review flagged media.
  - Provides context-sensitive judgment (e.g., journalistic ethics, medical confidentiality, legal evidence).

- Continuous Adaptation & Feedback Loop
  - New deepfake samples are continuously added to training datasets.
  - Models are updated regularly to stay ahead of evolving attack methods.
  - Human reviewer feedback helps refine AI detection accuracy.

Ideal Anti-Deepfake

- Real-Time Detection, e.g., Lightweight Models (MesoNet)
  - All incoming media (uploads, shared content) is scanned first with fast, lightweight models like MesoNet.
    - quickly flag obviously suspicious material without heavy computation.
  - Helps filter large volumes of content (e.g., on social media).

- Advanced Analysis, e.g., Ensemble & Vision Transformer (ViT)
  - Content flagged by Layer 2 goes through more computationally intensive detection:
    - EnsembleNet → balances multiple detectors for robustness.
    - ViT → captures global anomalies (lighting, context, identity consistency) that CNNs may miss.
  - Critical for high-stakes scenarios (e.g., hospital patient identity verification, court evidence, election content).

Ideal Anti-Deepfake

- No single method is sufficient.
- The strongest defense is a multi-layered strategy:
  - Prevention (watermarking) → trust at the source.
  - Detection (MesoNet, EnsembleNet, ViT) → scalable + accurate filtering.
  - Human oversight → final safeguard.